

Nivell 1

Crea una base de dades amb MongoDB utilitzant com a col·leccions els arxius adjunts.

Exercici 1

- Mostra els 2 primers comentaris que hi ha en la base de dades.

The screenshot shows the MongoDB Compass interface. At the top, there's a query editor with a JSON query: `{}`. Below the query, there are tabs for **Project**, **Sort**, **Collation**, and **Index Hint**. The **Sort** tab is selected, showing `{ _id: 1 }`. To the right, there are input fields for **Max Time MS** (60000), **Skip** (0), and **Limit** (2). Below the query editor, there are buttons for **ADD DATA**, **UPDATE**, **DELETE**, **EXPORT DATA**, and **EXPORT CODE**. The results section shows two documents:

```
{
  "_id": ObjectId('5a9427648b0beeb69579cc'),
  "name": "Andrea Le",
  "email": "andrea_le@fakegmail.com",
  "movie_id": ObjectId('573a1390f29313caab418c'),
  "text": "Rem officiiis eaque repellendus amet eos doloribus. Porro dolor volupta...",
  "date": 2012-03-26T23:20:16.000+00:00
}
```

```
{
  "_id": ObjectId('5a9427648b0beeb69579cf'),
  "name": "Greg Powell",
  "email": "greg_powell@fakegmail.com",
  "movie_id": ObjectId('573a1390f29313caab41b1'),
  "text": "Tenetur dolorum molestiae ea. Eligendi praesentium unde quod porro. Co...",
  "date": 1987-02-10T00:29:36.000+00:00
}
```

- Quants usuaris tenim registrats?

The screenshot shows the MongoDB Compass interface with the **Aggregations** tab selected. The top bar shows **Documents** (185), **Aggregations**, **Schema**, **Indexes** (1), and **Validation**. Below the tabs, there's a query editor with a placeholder query: `{ field: 'value' }`. To the right, there are buttons for **Explain**, **Reset**, **Find**, and **Options**. Below the query editor, there are buttons for **ADD DATA**, **UPDATE**, **DELETE**, **EXPORT DATA**, and **EXPORT CODE**. The results section shows a preview of the aggregation pipeline.

Sabemos que en la colección users, cada documento pertenece a un usuario registrado.

- Quants cinemes hi ha en l'estat de Califòrnia?

The screenshot shows the MongoDB Compass interface with the **Aggregations** tab selected. The top bar shows **Documents** (1.4K), **Aggregations**, **Schema**, **Indexes** (1), and **Validation**. Below the tabs, there's a query editor with a placeholder query: `{ field: 'value' }`. To the right, there are buttons for **Explain**, **Run**, and **Options**. Below the query editor, there are buttons for **SAVE**, **CREATE NEW**, **EXPORT DATA**, and **EXPORT CODE**. The results section shows a preview of the aggregation pipeline:

```
Stage 1: $match
1 { "location.address.state": "CA" }
```

Output preview after \$match stage (Sample of 10 documents):

```
{
  "_id": ObjectId('59a47286cfa9a3a73e51e72e'),
  "theaterId": 1008,
  "location": Object
}
```

```
Stage 2: $count
1 /**
2  * Provide the field name for the count:
3  */
4 "total_theaters_CA"
```

Output preview after \$count stage (Sample of 1 document):

```
{
  "total_theaters_CA": 169
}
```

○ Quin va ser el primer usuari/ària en registrar-se?

The screenshot shows the MongoDB Atlas interface for a database named 'cinema_app'. The 'users' collection is selected. The query editor shows a simple query: `{}`. The 'Sort' field is set to `{ _id: 1 }`. The 'Limit' is set to 1. The 'Find' button is highlighted. Below the query editor, the first document is displayed:

```
{
  "_id": ObjectId("59b99db4cfa9a34dcd7885b6"),
  "name": "Ned Stark",
  "email": "sean_bean@gameofthron.es",
  "password": "$2b$12$UREFwsRUoyF0CRqGNK0Lz00HM/jLhgUCNNI39RJAqMUQ74cr131Vu"
}
```

El ObjectId de MongoDB contiene el timestamp de creación, por lo que ordenar por `_id: 1` devuelve el primer documento insertado (usuario registrado primero).

○ Quantes pel·lícules de comèdia hi ha en la nostra base de dades?

The screenshot shows the MongoDB Atlas interface for the 'cinema_app' database, specifically the 'movies' collection. An aggregation pipeline is defined with two stages:

- Stage 1:** `$match: { genres: "Comedy" }`. The output preview shows a sample of 10 documents, including details like `_id`, `plot`, `genres`, `runtime`, `cast`, `num_mflix_comments`, and `poster`.
- Stage 2:** `$count: "total_comedy_movies"`. The output preview shows a single document with the result: `total_comedy_movies: 7624`.

Exercici 2

Mostra'm tots els documents de les pel·lícules produïdes en 1932, però que el gènere sigui drama o estiguin en francès.

The screenshot shows the MongoDB Compass interface. The top navigation bar includes 'Welcome', 'My Queries', 'Data Modeling', 'Nizar', 'users', 'comments', 'movies', 'sessions', 'theaters', and a plus icon. The main header shows 'Nizar > cinema_app > movies' and a button to 'Open MongoDB shell'. Below the header, there are tabs for 'Documents' (24K), 'Aggregations', 'Schema', 'Indexes' (1), and 'Validation'. The 'Documents' tab is active, displaying a query builder interface. The query is:

```
{
  year: 1932,
  $or: [
    { genres: "Drama" },
    { languages: "French" }
  ]
}
```

 The 'Project' field is set to '{ field: 0 }'. The 'Sort' field is set to '{ field: -1 } or [[\'field\', -1]]'. The 'Collation' field is set to '{ locale: \'simple\' }'. The 'Index Hint' field is set to '{ field: -1 } or "indexName"'. The 'Max Time MS' is set to 60000. The 'Skip' and 'Limit' fields are both set to 0. Below the query builder, there are buttons for 'ADD DATA', 'UPDATE', 'DELETE', 'EXPORT DATA', and 'EXPORT CODE'. The bottom section shows the JSON document for the movie 'The Blood of a Poet'.

```
{
  "_id": ObjectId("573a1391f29313caabed9458"),
  "plot": "A young artist draws a face at a canvas on his easel. Suddenly the mou...",
  "runtime": 55,
  "rated": "UNRATED",
  "cast": Array (4),
  "num_flix_comments": 1,
  "poster": "https://m.media-amazon.com/images/M/MV5BYWY3ODk5ZWYtYjlmYi00NjA4LTk4Zm...",
  "title": "The Blood of a Poet",
  "lastupdated": "2015-09-16 13:13:05.537000000",
  "languages": Array (1),
  "released": 2010-05-20T00:00:00.000+00:00,
  "directors": Array (1),
  "writers": Array (1),
  "awards": Object
}
```

Exercici 3

Mostra'm tots els documents de pel·lícules estatunidenques que tinguin entre 5 i 9 premis que van ser produïdes entre 2012 i 2014.

The screenshot shows the MongoDB Compass interface. The top navigation bar includes 'Welcome', 'My Queries', 'Data Modeling', 'Nizar', 'users', 'comments', 'movies', 'sessions', 'theaters', and a plus icon. The main header shows 'Nizar > cinema_app > movies' and a button to 'Open MongoDB shell'. Below the header, there are tabs for 'Documents' (24K), 'Aggregations', 'Schema', 'Indexes' (1), and 'Validation'. The 'Documents' tab is active, displaying a query builder interface. The query is:

```
{
  countries: "USA",
  "awards.wins": { $gte: 5, $lte: 9 },
  year: { $gte: 2012, $lte: 2014 }
}
```

 The 'Project' field is set to '{ field: 0 }'. The 'Sort' field is set to '{ field: -1 } or [[\'field\', -1]]'. The 'Collation' field is set to '{ locale: \'simple\' }'. The 'Index Hint' field is set to '{ field: -1 } or "indexName"'. The 'Max Time MS' is set to 60000. The 'Skip' and 'Limit' fields are both set to 0. Below the query builder, there are buttons for 'ADD DATA', 'UPDATE', 'DELETE', 'EXPORT DATA', and 'EXPORT CODE'. The bottom section shows the JSON document for the movie 'The Secret Life of Walter Mitty'.

```
{
  "_id": ObjectId("573a13acf29313caabd29366"),
  "fullplot": "The manager of the negative assets sector of Life magazine, Walter Mit...",
  "imdb": Object,
  "year": 2013,
  "plot": "When his job along with that of his co-worker are threatened, Walter t...",
  "genres": Array (3),
  "rated": "PG",
  "metacritic": 54,
  "title": "The Secret Life of Walter Mitty",
  "lastupdated": "2015-08-31 00:10:51.747000000",
  "languages": Array (3),
  "writers": Array (3),
  "type": "movie",
  "tomatoes": Object,
  "poster": "https://m.media-amazon.com/images/M/MV5BODYwNDYxNDk1N1S0M1S0anBxkFzT...",
  "num_flix_comments": 1
}
```

Nivell 2

Exercici 1

Compte quants comentaris escriu un usuari/ària que utilitza "GAMEOFTHRON.ES" com a domini de correu electrònic.

The screenshot shows the MongoDB Atlas interface for a database named 'cinema_app'. The 'Aggregations' tab is selected, and a pipeline with two stages is defined:

- Stage 1: \$match** with the query: `{ "email": "@gameofthron\\.es$", $options: "i" } }`. The output preview shows two sample documents with email addresses from 'gameofthron.es'.
- Stage 2: \$count** with the query: `"total_comments_gameofthrones_dom"`. The output preview shows a single document: `{ "total_comments_gameofthrones_dom": 22841 }`.

Exercici 2

Quants cinemes hi ha en cada codi postal situats dins de l'estat Washington D. C. (DC)?

The screenshot shows the MongoDB Atlas interface for a database named 'cinema_app'. The 'Aggregations' tab is selected, and a pipeline with three stages is defined:

- Stage 1: \$match** (empty query).
- Stage 2: \$group** with the query: `{ "_id": "$location.address.zipcode", "total_theaters": { $sum: 1 } }`. The output preview shows three sample documents with zip codes and their theater counts.
- Stage 3: \$sort** with the query: `{ "total_theaters": -1 }`. The output preview shows the same three documents, sorted by theater count in descending order.

Nivell 3

Exercici 1

Troba totes les pel·lícules dirigides per John Landis amb una puntuació IMDb (Internet Movie Database) d'entre 7,5 i 8.

WelcomeMy QueriesData ModelingNizaruserscommentsmoviessessions theaters+

Nizar > cinema_app > movies

Documents 24KAggregationsSchemaIndexes 1Validation

Generate query +

Explain

Reset

Find

Options

Project

{ field: 0 }

Sort

{ field: -1 } or [['field', -1]]

Collation

{ locale: 'simple' }

Index Hint

{ field: -1 } or "indexName"

Max Time MS

60000

Skip

0

Limit

0

ADD DATA

UPDATE

DELETE

EXPORT DATA

EXPORT CODE

25

1 - 4 of 4

_id: ObjectId('573a1397f29313caabce6d94')

fullplot: "Faber College has one frat house so disreputable it will take anyone. ..."

imdb: Object

year: 1978

plot: "At a 1962 college, Dean Vernon Wormer is determined to expel the entire..."

genres: Array (1)

rated: "R"

metacritic: 82

title: "Animal House"

lastupdated: "2015-09-13 00:02:47.803000000"

languages: Array (2)

writers: Array (3)

type: "movie"

tomatoes: Object

poster: "https://m.media-amazon.com/images/M/MV5BM2M2ZDA4NTYtOGZjM108OTg5LWI1ZT..."

num_reflix_comments: 1

released: 1978-07-28T00:00:00.000+00:00

awards: Object

countries: Array (1)

cast: Array (4)

directors: Array (1)

runtime: 109

Exercici 2

Mostra en un mapa la ubicació de tots els teatres de la base de dades.

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